IST 691: Research Methodology and Scientific Writing Skills)

Control in Experimentation

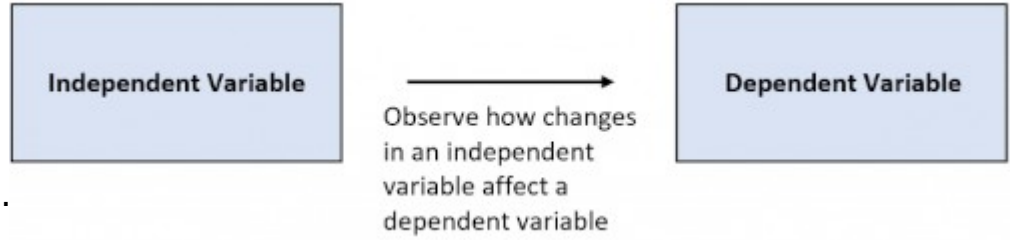
Denis L. Nkweteyim

Outline

- Introduction
- Control of Extraneous Variables
- Techniques for Achieving Constancy

Introduction

- A major goal in any experiment is to attain internal validity
- Internal validity
  - The extent to which we can accurately state that the independent variable produced the observed effect
    - i.e., the extent to which the observed effect is caused only by the experimental treatment condition

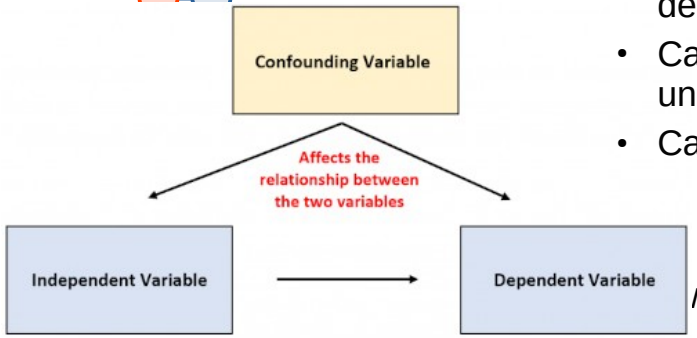


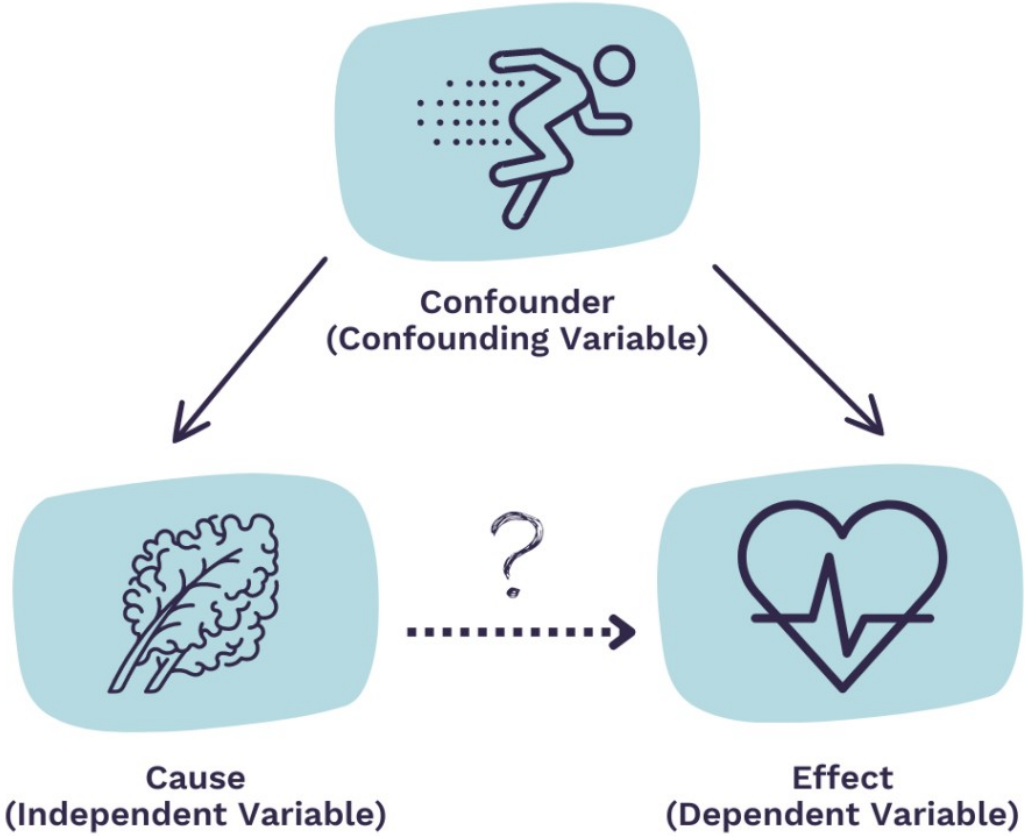
Introduction

- The dependent variable could have been influenced by variables other than the independent variable
- Example
  - Investigating the effect of tutoring (independent variable) on grades (dependent variable)
  - We want to be able to conclude that any improvement in grades of students who received tutoring over those who did not was as a result of the tutoring
  - But what if tutored students were brighter than those who were not tutored
    - The improvement in grades could be due to the fact that the tutored students were brighter
    - Intelligence here represents an extraneous variable

Introduction

- Extraneous variable
  - Any variable other than the independent variable that influences the dependent variable
  - Confounding
    - Occurs when the experiment contains a variable that systematically varies with variations in the independent variable
    - A confounding variable
      - A variable that is not included in an experiment, yet affects the relationship between the dependent and independent variables
      - Can affect both the independent variable and the dependent variable, causing unexpected results in the study
      - Can confound the results of an experiment and lead to unreliable findings




Confounder
(Confounding Variable)
?
Cause
(Independent Variable)
Effect
(Dependent Variable)

Confounding Variable

- Key Characteristics
  - Correlated with independent variable, i.e., the factor being studied (e.g., coffee drinking)
  - Causally related to dependent Variable by affecting the outcome (e.g., heart disease).
- Example: Coffee & Heart Disease
  - Hypothesis: Coffee causes heart disease
  - Independent Variable: Coffee consumption
  - Dependent Variable: Heart disease
  - Confounding Variable: Smoking
    - Smokers often drink more coffee and smoking causes heart disease
    - The confounder (smoking) makes it seem like coffee causes heart disease, when smoking is the real culprit

Confounding Variable

- Why They Matter
  - The distort Results: By hiding or exaggerating the true effect of the independent variable
  - They may invalidate conclusions: You might draw incorrect conclusions about cause and effect.
- How to Control Them
- Randomization: Randomly assigning participants to groups helps distribute confounders evenly.
- Blinding: Single or double-blinding (neither participant nor researcher knows treatment) prevents bias
- Stratification: Grouping data by the confounder to analyze effects within each group
- Statistical Adjustment: Using methods to statistically account for the confounder's effect

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Control of Extraneous Variables

- The effects of some extraneous variables can be eliminated especially in a laboratory setting
- But there are usually several extraneous variables whose effect cannot be eliminated
- Although it is not possible to eliminate the influence of such variables from the experiments
  - It is possible to eliminate the differential influence that these variables may have across the various levels of the independent variable
  - i.e., it is possible to keep the influence of these variables constant across the various levels of the independent variable
- Constancy
  - Stability, or absence of change, in the influence exerted by an extraneous variable in all treatment conditions

Extraneous Variables to be Controlled

- History
  - External events occurring during the study that affect participants, unrelated to the treatment, like a new public health campaign influencing a diet study
- Maturation
  - Natural, internal changes within participants over time (e.g., aging, fatigue, learning) that might alter behavior, independent of the study
  - e.g., if the project lasts a few years, most participants may improve their performance regardless of treatment
- Testing
  - The effect of taking a pre-test, which can influence scores on a post-test, making participants more familiar with the material or sensitized to the test
- Instrumentation
  - Changes in the measurement tool or procedure (e.g., a worn-out scale, different interviewers) between measurements, altering what is being assessed

Extraneous Variables to be Controlled

- Statistical regression
  - Also known as regression to the mean
  - The tendency of extreme pre-test scores to revert back toward the population mean, such that when individuals are selected for program participation based on extreme pretest results their posttest scores will tend to shift toward the mean score, regardless of the efficacy of the program
  - Threat: If you select participants with extreme scores (e.g., lowest performers for a new program), their improvement might just be regression, not the program's effect
- Selection of subjects (Selection Bias)
  - What it is: Differences between groups that exist before the experiment starts, often due to non-random assignment.
  - Threat: If one group is naturally more motivated or capable, any outcome differences might be due to selection, not the treatment.
  - Control: Random assignment and matching participants across groups help equalize these pre-existing differences
- Experimental mortality
  - What it is: Participants dropping out of a study, especially if it's unequal across groups
  - Threat: The remaining participants might not represent the original group, and those who stay might be different (e.g., more motivated)
  - Example: In a long-term study, motivated students finish, skewing results to look better than if everyone had stayed

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Techniques for Achieving Constancy

- Randomization
- Matching
- Counterbalancing
- Control of Participant Effects
- Control of Experimental Effects

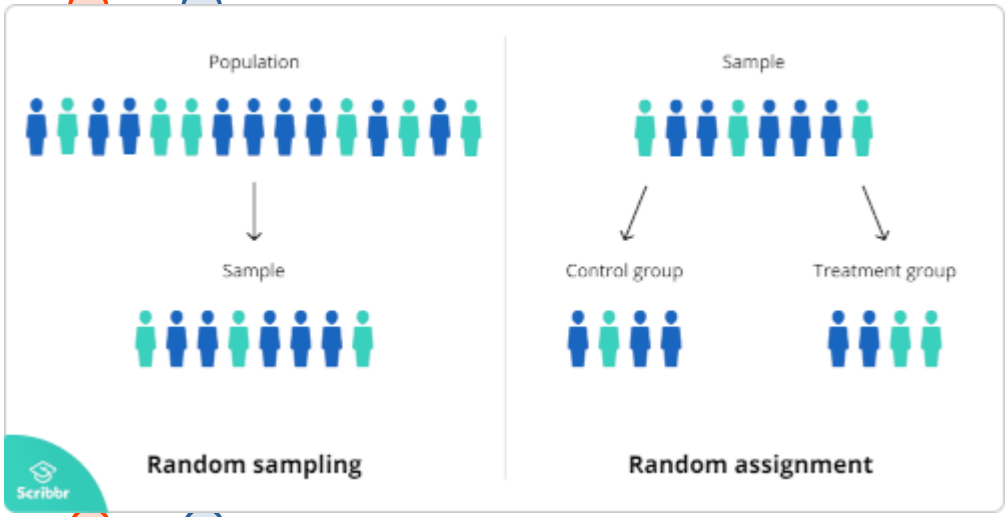
Randomization

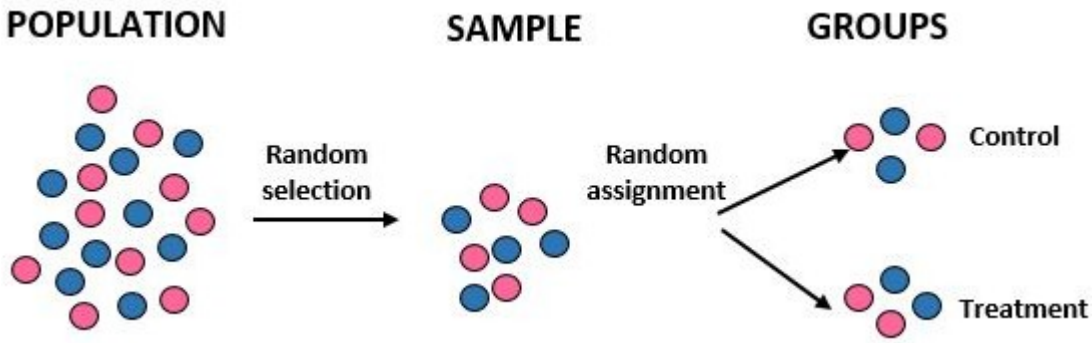
- A control technique that equates groups of participants by ensuring every member has an equal chance of being assigned to any group
- Randomly selecting research participants from a population provides maximum assurance that a systematic bias does not exist in the selection process and that the selected sample is representative of the population
- Representative sample
  - Sample participants have the same characteristic as the people in the population

Randomization

- Selecting a representative sample
  - Can only be done by selecting a random sample
  - Random sample
    - A sample of participants that is selected in such a way that each person has an equal chance of being selected and the selection of one does not affect the selection of another
  - Random assignment
    - The randomly selected participants should be randomly assigned to the same number of groups as there are experimental treatment conditions

Randomization





Randomization

- Random assignment
  - Provides maximum insurance that groups are equal
  - Eliminates systematic differences between groups
  - Does not eliminate extraneous variables, but randomly distributes them across groups

Various Flavours of Randomization

- Simple Random Sampling
- Stratified random sampling
- Systematic random sampling
- Cluster sampling

Simple Random Sampling

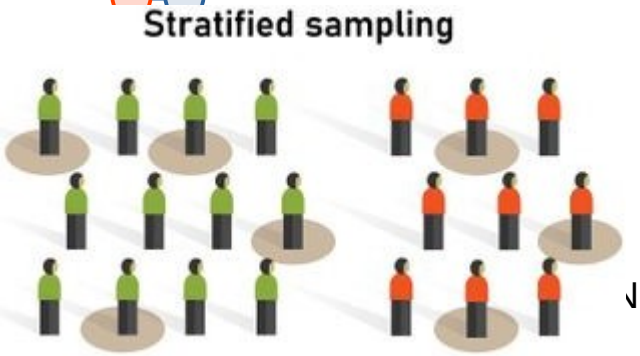
- Each member of population is equally likely to be chosen as part of the sample
- Purest and the most straightforward probability sampling strategy
- Considered to be the most unbiased representation of population

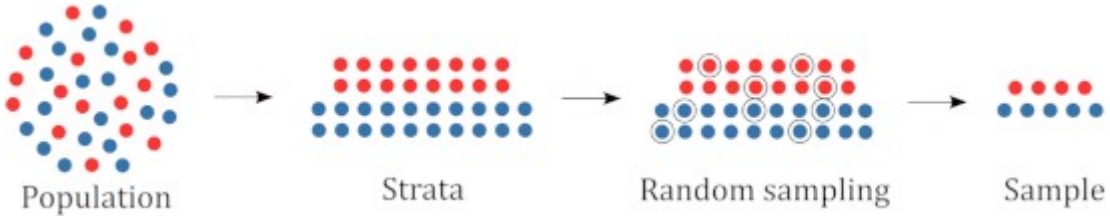
Simple Random Sample Generation Example

- Say, your population are employees of company ABC Limited that has 600 employees
- You can apply simple random sampling as follows
  - Prepare the list of all 600 employees working for ABC Limited
  - Assign a sequential number for each employee from 1 to N (in your case from 1 to 600)
  - Determine the sample size (assume sample size of 150 participants)
  - Use random number generator software to generate 150 numbers from 1 to 600
  - The 150 employees assigned with respective generated numbers are going to represent sample group members for your research

Stratified Random Sampling

- Probability sampling method and a form of random sampling
- Objective
  - To guarantee that the sample represents specific sub-groups or strata
- Method involves dividing population into different subgroups (strata) according to one or more common attributes
  - Sex, age, income, level of education etc. according to aims and objectives of the study
- Then selecting subjects from each strata in a proportionate manner.





Cluster Sampling

- Used when natural groups are present in a population
  - The whole population is subdivided into clusters
  - A few of the available clusters are chosen randomly and the others eliminated
- Clusters consist of internally heterogeneous and externally homogeneous subpopulations
  - Externally homogeneous as they appear to be grouped together by a shared characteristic/criteria
  - Internally heterogeneous because the subpopulations within the clusters have different compositions
- Examples
  - Clusters may be divided by different cities in a country, different areas in a city, different organizations, different universities
- Cluster sampling process can be single stage
  - All members of selected clusters are included in the study, or
- Multistage
  - Additional sampling methods are used to choose certain individuals within selected clusters

Example: Assigning Participants to Groups

- Consider a study on learning
- We know that intelligence is correlated to learning, so it needs to be controlled in the experiment
- Consider 2 possibilities
  - One that provides for the needed control through random assignment, and
  - The other that does not
    - First 10 participants assigned to Group A and the next 10 assigned to Group B

Example: Assigning Participants to Groups

Hypothetical Distribution of 20 Research Participants' IQ Scores

Group Assignment Based on Arrival Sequence				Random Assignment of Participants to Groups			
Group A		*Group B*		*Group A*		*Group B*	
Participants	*IQ Scores*	*Participants*	*IQ Scores*	*Participants*	*IQ Scores*	*Participants*	*IQ Scores*
1	97	11	100	1	97	3	100
2	97	12	108	2	97	4	103
3	100	13	110	11	100	6	108
4	103	14	113	5	105	12	108
5	105	15	117	13	110	7	109
6	108	16	119	9	113	8	111
7	109	17	120	15	117	14	113
8	111	18	122	10	118	16	119
9	113	19	128	19	128	17	120
10	118	20	130	20	130	18	122
Mean IQ score	106.1		116.7		111.5		111.3
Difference between Group A mean and Group B mean: 10.6				Difference between Group A mean and Group B mean: .2			

Example: Assigning Participants to Groups

- Assume that from the results of the experiment when participants were assigned to groups based on when they arrived
  - Treatment Group B learned significantly faster than Treatment Group A
  - Is this difference caused by the different experimental treatment conditions that were administered to the 2 groups, or by the fact that participants in Group B may have been more intelligent than those in Group A (as suggested by the superior average IQ scores for Group B participants)
  - We can’t draw that conclusion because intelligence is a potential confounding variable, and serves as a rival hypothesis for explaining the observed performance

Example: Assigning Participants to Groups

- Now, assume that participants are randomly assigned to the two treatment groups
  - This time, the difference in the average IQ scores is much smaller (0.2 vs 1.6)
    - Both groups have similar IQ scores, the effect of which is to control for the potential biasing effect of IQ
- Random assignment produces control because the variables to be controlled are distributed in approximately the same manner in all the groups, thus holding the effect of the extraneous variable constant

Example: Assigning Participants to Groups

- Does randomization always result in equal distribution of the variables to be controlled?
- No. Although randomization ensures that the extraneous variables to be controlled are distributed randomly, it is still possible that by chance, these variables are not distributed equally among the groups
- And then, there are some extraneous variables that are not controlled by randomization
  - So, other techniques need to be used together with randomization, to control these variables
    - Matching, Counterbalancing, Control of participant effects, Control of experimenter effects

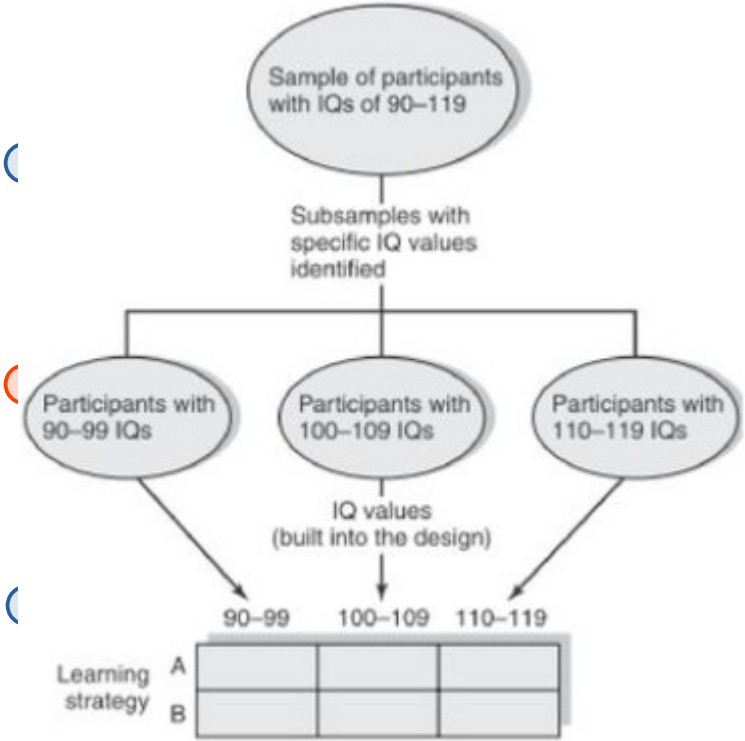
Techniques for Achieving Constancy

- Randomization
- Matching
- Counterbalancing
- Control of Participant Effects
- Control of Experimental Effects

Matching

- Use of any of a variety of techniques for equating participants in the treatment groups on specific variables
  - Should be done with variables thought to be related to the independent variable, or may confound the independent variable
- Advantages
  - Controls for the variables on which participants are matched
  - Increases the sensitivity of the experiment

Matching by Building the Extraneous Variable into the Research Design (Blocking)

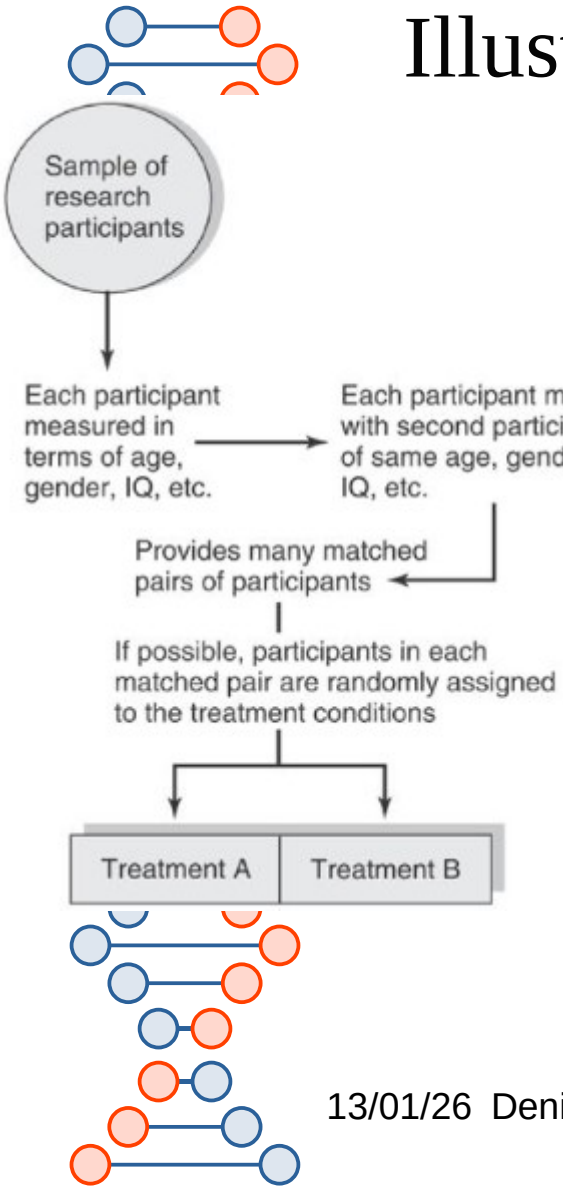


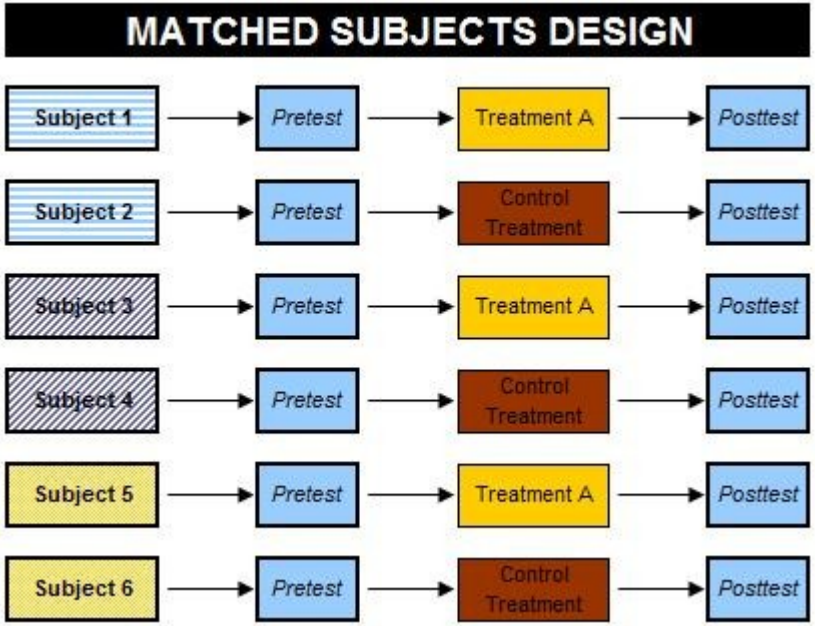
- Here, we convert the extraneous variable into another independent variable
  - e.g., Dependent variable scores on the different IQ levels in the design on the left
- Use only when you are interested in the effect of the extraneous variable

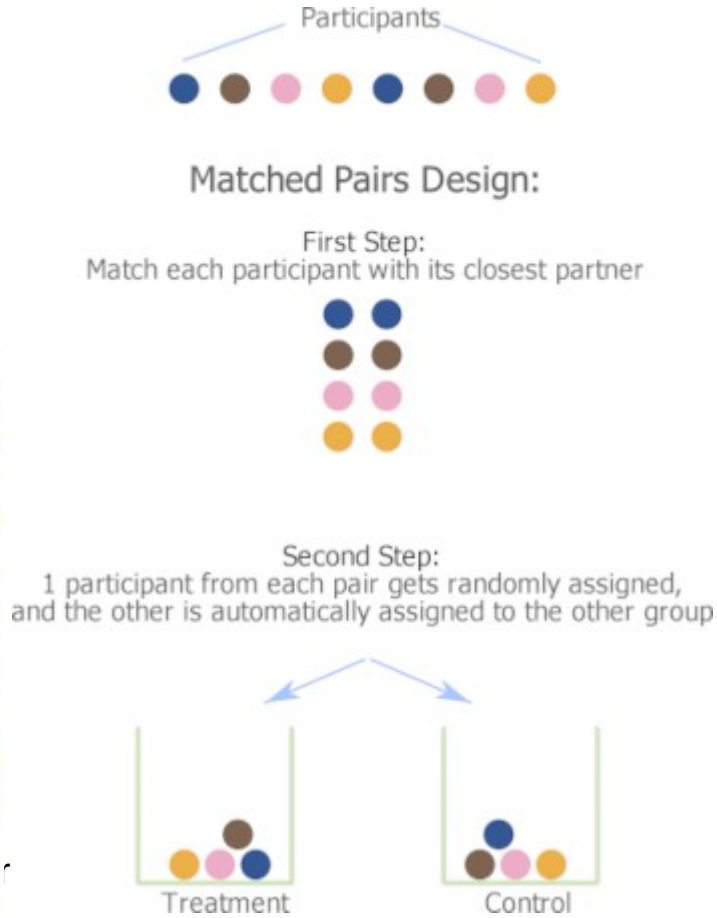
Matching by Equating Participants

- Similar to matching by building the extraneous variable into the design
  - Both techniques attempt to eliminate the the influence of the extraneous variable by creating equivalent groups of participants
  - Difference lies in the procedure for creating the equivalent groups
    - Previous method: new independent variable created based on categories of the extraneous variable
    - Present method: matches participants to the variable to be controlled, where the number of participants is always some multiple of the number of levels the independent variable

Illustrating Matching by Equating Participants







Individual Matching Example

Name	IQ
Sara	110
Jane	95
Donald	120
Debbie	105
Jake	120
Jody	105
Kathy	95
Brian	110

Random Assignment to conditions:

Treatment	Control
Brian	Sara
Jane	Kathy
Debbie	Jody
Jake	Donald

Average IQ by group:

Treatment	Control
107.5	107.5

Techniques for Achieving Constancy

- Randomization
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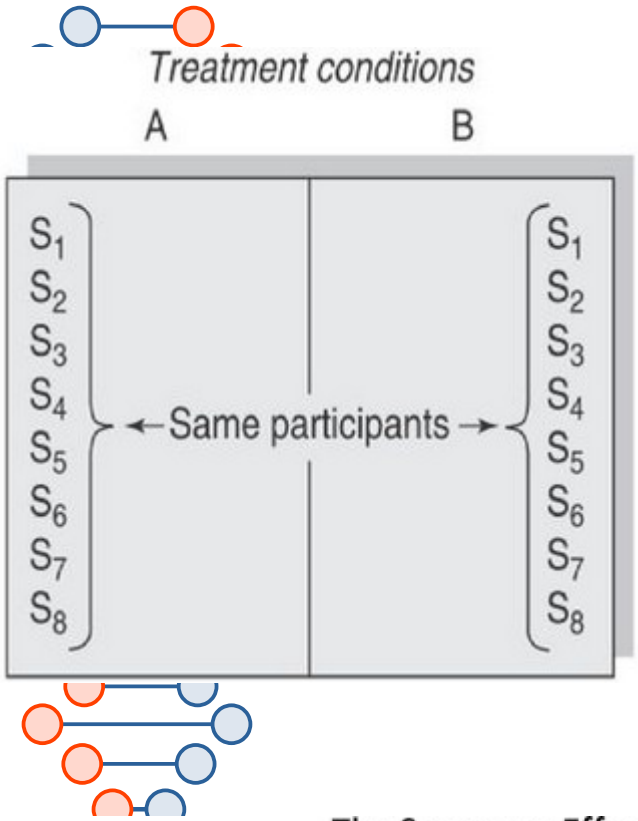
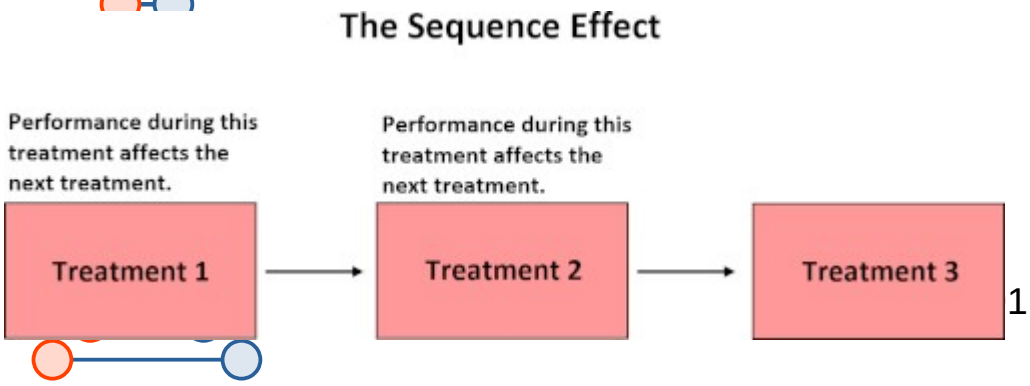


Counterbalancing

- Technique used to control sequencing effects
- Sequencing effects
  - Occur when the sequence of experimental treatments given to participants in a research study interact with each other
    - Can occur when the researcher elects to construct an experiment in which all participants serve in each of the experimental conditions



Two Types of Sequencing Effects

- Order effects
- Carry-over effects

Order Effects

- Sequencing effect arising from the order in which the conditions are administered to participants
  - e.g., practice, fatigue

Practice Effect

Participants perform better during later treatment conditions because they've had time to practice and improve.

Fatigue Effect

Participants perform worse during later treatment conditions because they're tired or fatigued from previous treatment conditions.

Carry-over Effects

- Sequencing effect that occurs when performance in one treatment condition affects performance in another treatment condition
  - e.g., effect of Treatment A still affecting participant when engaged in Treatment B

Counterbalancing Procedures

- Randomized counterbalancing
  - Sequence of conditions is randomly determined for each participant
  - Example
    - Independent variable with three levels, ABC
    - Six possible sequence (ABC, ACB, BAC, BCA, CAB, CBA)
    - Each participant randomly assigned to one sequence

Counterbalancing Procedures

- Intra-subject counterbalancing: The ABBA technique
  - Each participant takes the treatment conditions first in one order and then in the reverse order
  - Example: experiment to determine if people prefer Pepsi Cola (Cola A) or Coke (Cola B)
  - Treatment conditions: Cola A and Cola B
    - Taste Cola A, then Cola B, then make an assessment of each
    - Then taste Cola B and then Cola A and make an assessment, making the sequence ABBA
- But there could still be a differential order effect
  - So also create another sequence: BAAB
  - One group of participants follow sequence ABBA, another group sequence BAAB

Counterbalancing Procedures

- Intra-group counterbalancing
  - Primary disadvantage of intra-subject counterbalancing
    - Each treatment condition must be presented to each research participant more than once
    - As the number of treatment conditions increases, so too the length of the sequence of conditions each participant must take
      - e.g., 3 treatment conditions ==> six treatment conditions for each participant: ABC, ACB, …, CBA
  - Intra-group counterbalancing
    - Groups of participants, and not individual participants, are counterbalanced

Incomplete counterbalancing

- Intra-group counterbalancing in which not all the possible sequences of treatment conditions are enumerated
  - This is the most frequently used form of intra-group counterbalancing
- Consider a study to determine whether the amount of caffeine affects reaction time
  - Four dosages of caffeine (100, 200, 300, 400 mg) corresponding to conditions A, B, C, D administered to participants and reaction times measured
  - Sequencing effect will probably occur if each participant was given all four doses
    - So, counterbalance the order in which the dosages are administered

Incomplete counterbalancing

- Could adopt the following strategy
  - Whenever the number of treatment conditions is even, make the number of counterbalanced sequences equal to the number of treatment conditions
  - Establish sequences as follows
    - First sequence takes the form 1,2, n, 3, (n-1), 4, (n-2), 5, …
      - For the four-dosages: ABDC (i.e., 1, 2, 4, 3)
      - If there were 6 treatment conditions: ABFCED (i.e., 1, 2, 6, 3, 5, 4)
    - The remaining sequences of the incomplete counterbalancing technique established by incrementing each value in the preceding sequence by 1
  - Note
    - Each value precedes and follows every other value the same number of times (1 time in this example)

Participant	Sequence			
1	A	B	D	C
2	B	C	A	D
3	C	D	B	A
4	D	A	C	B

Incomplete counterbalancing

- What if the number of treatment conditions is odd (5 conditions say)
- The condition that each value must precede and follow every other value the same number of times is not fulfilled
  - Although each treatment condition appears in every possible position, A for example is immediately precede by C twice, but never by B
  - Remedy the situation by adding five additional sequences that are exactly the reverse of the first five sequences

Sequence				
D	C	E	B	A
E	D	A	C	B
A	E	B	D	C
B	A	C	E	D
C	B	D	A	E

Sequence				
A	B	E	C	D
B	C	A	D	E
C	D	B	E	A
D	E	C	A	B
E	A	D	B	C

Techniques for Achieving Constancy

- Randomization
- Matching
- Counterbalancing
- Control of Participant Effects
- Control of Experimenter (Researcher) Effects

Control of Participant Effects

- Participants' behaviour in an experiment can be influenced by perceptions and motives they bring with them
  - They tend to be motivated to present themselves in the best possible light
- We consider two ways to control participant effects
  - Double blind placebo model
  - Deception

Double Blind Placebo Model

- One of the best techniques for controlling participant (and experimenter) effects
  - Neither the experimenter nor the research participant is aware of the treatment condition administered to the participant
    - The experimenter – like the participant – does not know which group receives the placebo condition and which group receives the experimental manipulation

Deception

- One of the most common methods to solve the problem of participant perceptions
- Giving the participant a bogus rationale for the experiment
  - Provide all research participants with a hypothesis that is unrelated to, or orthogonal to the real hypothesis
  - The risk of the false information must not outweigh the benefit of the information
  - Must keep the false information constant for all participants

Techniques for Achieving Constancy

- Randomization
- Matching
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- Control of Participant Effects
- Control of Experimenter Effects

Control of Experimenter Effects

- Experimenter effects
  - Unintentional biasing influence that can be exerted by the experimenter
- We consider 3 approaches to control experimenter effects
  - Control of recording errors
  - Control of attribute errors
  - Blind technique

Control of Recording Errors

- Make researchers aware of making careful observations
  - May require training of the researcher before data collection
- Use multiple data recorders (video cameras, computers) and observers
- Have participants make responses on a computer
  - Eliminates human recorder, and possible associated errors

Control Attribute Errors

- Refers to attributes of the experimenter
- Use the same experimenter in all treatment conditions, except the treatment conditions interact with experimenter attributes

Control of Experimenter Expectancies

- Blind technique
  - Method whereby knowledge of each research participant's treatment condition is kept from the experimenter
    - This is the experimenter half of the double blind placebo model
  - Experimenter knows the hypothesis but is blind as to which treatment condition the research participant is in
  - Consequently, the experimenter cannot unintentionally treat groups differently
- Partial blind technique
  - Used in cases in which the blind technique cannot be used
  - Knowledge of each research participant's treatment condition is kept from the experimenter through as many stages of the experiment as possible
- Automation
  - Totally automating the experimental procedures so that no experimenter-participant interaction is required
    - Instructions written, tape recorded, filmed, televised, or presented by means of a computer, and responses recorded via timers, counters, pen recorders, computers, or similar devices